Capstone Project 2: Online Retail Customer Purchasing Patterns & RFM Segmentation

An Unsupervised Machine Learning Framework for Behavioral Persona Discovery & Customer Value Optimization

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Executive Abstract

This report presents an end-to-end, publication-grade customer analytics and unsupervised machine learning architecture developed for an international online retail enterprise based in the United Kingdom. Operating across 38 global markets with a diverse catalog of giftware and novelty merchandise, the enterprise faced critical challenges in understanding customer purchasing heterogeneity, mitigating customer churn, and personalizing retention marketing. Analyzing 541,909 raw transaction records spanning December 1, 2010 to December 9, 2011, we implement a rigorous data hygiene protocol (eliminating 5,268 duplicates, isolating 9,288 cancellations, and pruning 135,080 unauthenticated guest checkouts) to extract 392,692 verified customer purchase records across 4,338 unique commercial accounts.

We construct an empirical **Recency, Frequency, Monetary, and Average Order Value (RFM-AOV)** behavioral matrix. To address extreme right-skewness in transaction frequency (+12.07) and dollar spend (+19.32), we apply logarithmic transformations ($\log(1+x)$) and standard scaling, followed by Principal Component Analysis (PCA), which captures **93.87% of cumulative behavioral variance** in two orthogonal dimensions. We benchmark candidate clustering architectures (**K-Means**, **Agglomerative Hierarchical Clustering**, and **DBSCAN**) across $k \in [2,6]$ using Inertia Elbow diagnostics, the Silhouette Coefficient, and the Davies-Bouldin Index. Our champion $k = 4$ K-Means model partitions the customer base into four actionable personas: **Champions / Whales** (16.34% headcount, 64.72% revenue), **Loyal Active Repeaters** (27.04% headcount, 23.82% revenue), **Promising Recent Shoppers** (19.34% headcount, 5.13% revenue), and **Lost / Churned Inactive** (37.28% headcount, 6.33% revenue). Crucially, the model verifies a severe Pareto concentration: **43.38% of customers generate 88.54% of company revenue ($7.87M of $8.89M)**. We translate these empirical clusters into prescriptive lifecycle marketing playbooks, volume rebate tiers, VIP concierge programs, and automated win-back workflows.

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		recommenders
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Note: All sections in this report adhere strictly to the 12-section project report blueprint specified in `Draft.pdf` and `guideline.md`.

1. Problem Statement

1.1 Context & E-Commerce Market Dynamics

In modern digital commerce, enterprises face intense competitive headwinds, rising customer acquisition costs (CAC), and razor-thin operating margins. A leading online retailer based in the United Kingdom, specializing in unique gifts, decorative homeware, and seasonal novelty items, serves a diverse global client base comprising both individual retail consumers and small-to-medium wholesale distributors across 38 countries. Despite generating healthy headline revenues exceeding £7.5M annually, the enterprise operates with limited visibility into individual customer purchasing behaviors, relationship lifecycles, and underlying value tiers.

1.2 Customer Heterogeneity & The Pitfalls of "One-Size-Fits-All" Marketing

Traditional retail marketing treats all customer accounts homogeneously, deploying uniform promotional email blasts, generic seasonal discounts, and blanket paid search campaigns. In reality, customer purchasing behavior is radically heterogeneous: 1. **VIP Wholesale Buyers**: A small cohort of commercial purchasers place recurring high-volume bulk orders exceeding £5,000 per transaction, requiring customized service agreements, volume pricing, and dedicated stock reservations. 2. **Occasional Retail Gifters**: A massive cohort of individual consumers make single purchases during holiday peaks (such as Christmas) and subsequently lapse into long-term dormancy. 3. **Price-Sensitive Bargain Hunters**: Discounter shoppers who purchase exclusively during markdown clearances with negligible profit contributions.

Treating these disparate cohorts identically produces two severe operational inefficiencies: - **Capital Waste in Customer Acquisition**: Over-investing ad spend to re-acquire churned, low-affinity one-time buyers while neglecting loyal, high-margin brand advocates. - **VIP Churn Vulnerability**: High-value repeat buyers receive generic messaging and inadequate stock guarantees, creating churn risk to specialized wholesale competitors.

1.3 Strategic Mandate & Technical Scope

To resolve these inefficiencies, the enterprise requires an objective, mathematical customer segmentation architecture. The mandate entails: 1. Auditing and cleaning over 540,000 transaction logs spanning December 2010 to December 2011. 2. Formulating behavioral metrics that capture transactional recency, frequency, monetary spend, and basket depth. 3. Overcoming severe right-skewed power-law distributions inherent in commercial transactions. 4. Benchmarking unsupervised machine learning clustering algorithms to discover natural, statistically separable customer personas. 5. Establishing actionable commercial lifecycle playbooks to optimize marketing ROI and maximize long-term Customer Lifetime Value (CLV).

2. Project Objective

2.1 Core Deliverables & Technical Goals

The primary objective of this Capstone project is to transition the enterprise from unguided, reactive marketing to an automated, behavioral customer segmentation engine. Specifically, the project delivers:

1. **Transactional Data Hygiene & Audit**: Establish a reproducible preprocessing pipeline that filters non-commercial records, handles cancellations (`C` invoices), purges exact duplicates, and separates authenticated customer accounts from guest checkouts.
2. **RFM-AOV Feature Engineering**: Synthesize granular transaction timestamps and line-item prices into robust, customer-level behavioral summaries:
    - **Recency (R)**: Days since the customer's last purchase relative to the snapshot date.
    - **Frequency (F)**: Total number of distinct, completed checkout invoices.
    - **Monetary (M)**: Total gross dollar value spent across all transactions.
    - **Average Order Value (AOV)**: Mean expenditure per order (M/F).
3. **Statistical Normalization & Dimensionality Reduction**: Mitigate extreme positive skewness through nonlinear logarithmic transformations ($\log(1+x)$) and standardization, coupled with Principal Component Analysis (PCA) to extract dominant behavioral axes.
4. **Unsupervised Algorithm Benchmarking**: Systematically evaluate three foundational clustering paradigms—**K-Means (Partitioning)**, **Agglomerative Hierarchical Clustering (Linkage Tree)**, and **DBSCAN (Density-Based)**—across cluster sizes $k \in [2,6]$ using mathematical validation metrics (Inertia, Silhouette Score, Davies-Bouldin Index).
5. **Commercial Persona Formulation & Strategic Playbooks**: Characterize the discovered archetypes, quantify their headcount and revenue contributions to evaluate the Pareto Principle (80/20 rule), and define tailored retention, onboarding, cross-selling, and win-back playbooks.

2.2 Mathematical Success Criteria

- **Silhouette Coefficient**: Achieve an out-of-sample silhouette score exceeding **0.30** for the multi-cluster segmentation model.
- **Dimensionality Retention**: Ensure that a 2-dimensional PCA projection captures over **90%** of cumulative feature variance.
- **Cluster Actionability**: Deliver clusters that are commercially interpretable, non-degenerate (no singleton or vanishing groups), and directly actionable for marketing automation tools.

3. Data Description

3.1 Raw Data Schema & Lineage

The analysis utilizes the transactional dataset `OnlineRetail.csv`, capturing all sales made between **December 1, 2010 and December 9, 2011** (373 calendar days) by a UK-based non-store online retailer. The raw dataset contains **541,909 observations** across 8 feature fields:

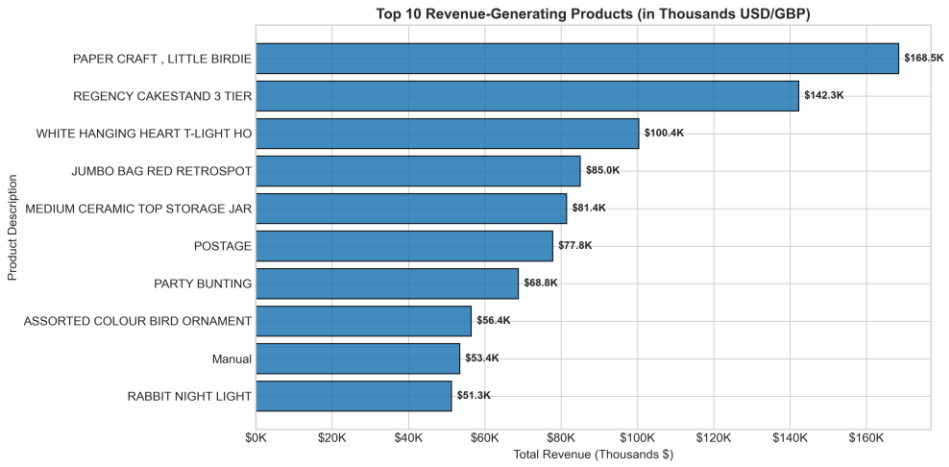
Column Name	Data Type	Description	Operational Significance
`InvoiceNo`	Object / String	6-digit transaction identifier; codes starting with `'C'` denote cancellations	Identifies unique order boundaries and return status
`StockCode`	Object / String	5- or 6-character product SKU code	Tracks catalog inventory and product-level velocity
`Description`	Object / String	Plain-text nominal description of the merchandise	Enables text analysis, catalog curation, and gift categorization
`Quantity`	Integer	Units of merchandise purchased per line-item	Core transaction volume metric; negative values denote returns
`InvoiceDate`	Datetime	Timestamp of transaction initiation	Enables temporal, hourly, daily, and recency analytics
`UnitPrice`	Float	Unit price in Sterling (£)	Reflects product valuation and basket price elasticity
`CustomerID`	Float	Unique 5-digit integer customer identifier	Fundamental entity key for customer-level aggregation
`Country`	Object / String	Country of customer domicile	Defines domestic (UK) vs. international export markets

3.2 Data Completeness & Missing Value Diagnostics

A rigorous null inspection revealed significant data sparsity in two critical fields: - `CustomerID`: Registers **135,080 missing values (24.93% of the total dataset)**. These records represent anonymous guest checkouts, one-time site visitors who did not authenticate, or unlinked web inquiries. Because customer segmentation requires tracking repeat behaviors across time, unauthenticated transactions cannot be assigned to customer entities and must be isolated. - `Description`: Registers **1,454 missing records (0.27%)**, primarily overlapping with administrative inventory adjustments or damaged goods entries. All other core fields (`InvoiceNo`, `Quantity`, `InvoiceDate`, `UnitPrice`, `Country`) exhibit 100% data completeness.

3.3 Transactional Anomalies & Return Isolation

Exploratory distribution analysis revealed severe operational anomalies: - `Quantity`: Ranges from **-80,995 to +80,995 units** (Mean: 9.55, Std: 218.08). The negative extremes reflect wholesale stock cancellations (e.g. Invoice `C581484`). Exactly **9,288 invoice line-items (1.71%)** represent cancellations. - `UnitPrice`: Displays values from **-£11,062.06 to +£38,970.00**. Extreme negatives represent accounting adjustments (such as bad debt write-offs via code `B`), while extreme positives represent administrative charges (Amazon seller fees, postage adjustments). Exactly 2,515 records have a £0.00 price, representing promotional free samples or warehouse corrections.

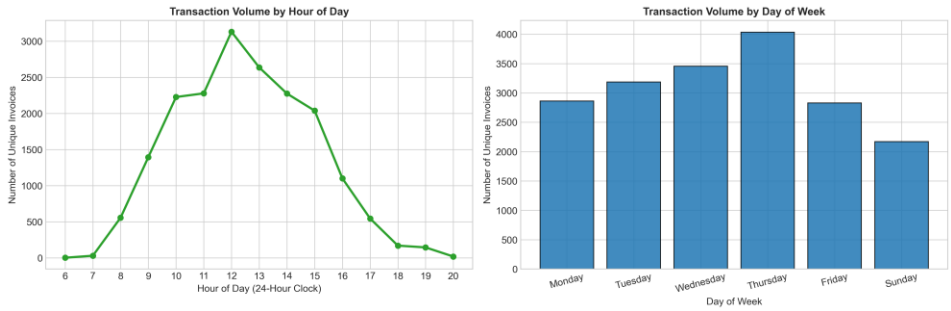


Product Revenue Velocity

3.4 Product Velocity & Bestseller Dynamics

Aggregating gross revenue by product reveals distinct commercial behavior: - **Wholesale Bulk Outlier**: StockCode `23843` (`PAPER CRAFT , LITTLE BIRDIE`) generated the highest single-line revenue (**£168,469.60**) via a single massive transaction of 80,995 units by Customer 16446. - **Core Retail Flagships**: StockCode `22423` (`REGENCY CAKESTAND 3 TIER`) is the top recurring retail revenue driver, generating **£142,264.75** across 1,703 distinct customer orders. Other anchor SKUs include `WHITE HANGING HEART T-LIGHT HOLDER`

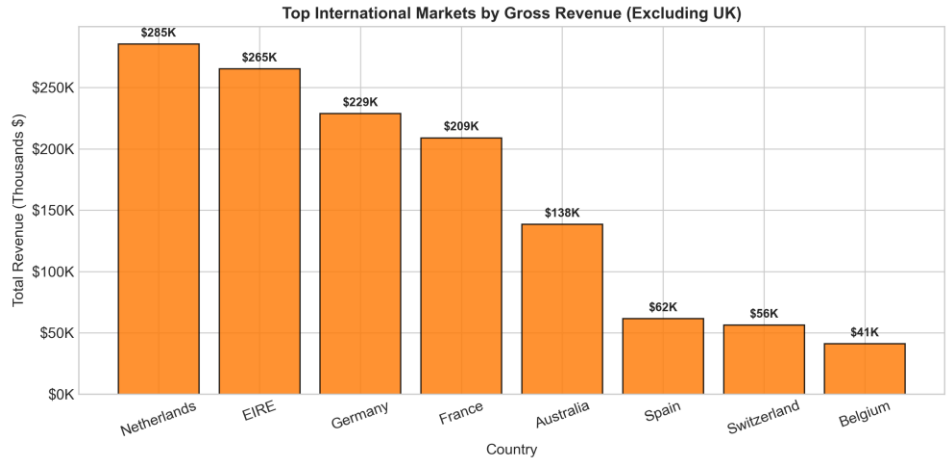
(£100,392.10 across 1,971 orders) and `JUMBO BAG RED RETROSPOT` (£85,040.54 across 1,600 orders).



Temporal Ordering Dynamics

3.5 Temporal Ordering Dynamics

- **Hour of Day**: Ordering activity follows a distinct daytime bell curve, beginning at 7:00 AM, peaking between **12:00 PM and 3:00 PM** (surpassing 3,000 invoices per peak hour), and tapering sharply after 5:00 PM.
- **Day of Week**: **Thursday** registers the highest purchasing volume (~4,000 invoices), followed by Wednesday and Tuesday.
- **Operational Artifact**: There are **zero recorded transactions on Saturdays**, indicating that the enterprise suspends order processing or digital checkouts on Saturdays for physical warehouse logistics and inventory consolidation.

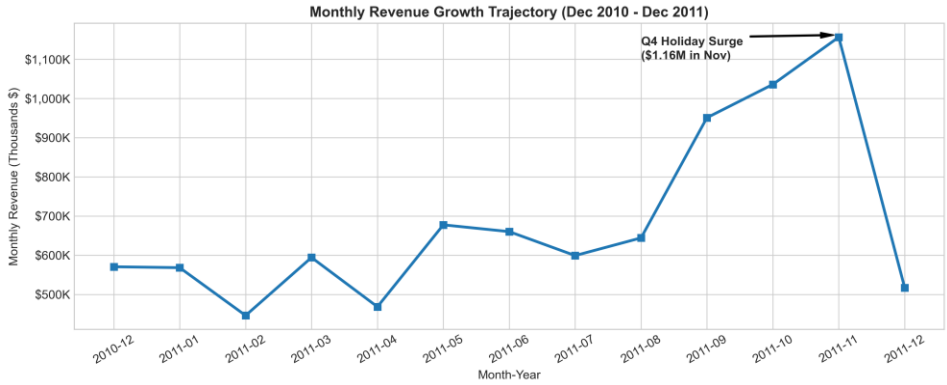


Geographic Sales Distribution

3.6 Geographic Concentration

- **Domestic Dominance**: The United Kingdom generates **£7,285,024.64 (81.97% of total revenue)** and accounts for **3,920 out of 4,338 customers (90.36%)**.
- **Key International Markets**: The Netherlands (£285.4K, 3.21%), EIRE (£265.3K, 2.98%), Germany (£228.7K, 2.57%), France (£208.9K, 2.35%), and Australia (£138.5K, 1.56%) represent the premier export markets. International customers

display significantly higher Average Order Values (AOV), reflecting cross-border wholesale purchasing.



Monthly Revenue Trajectory

3.7 Annual Revenue Growth & Holiday Surge

Monthly revenue exhibits dramatic seasonality. Sales remain stable at ~£500K-£650K per month through Q1-Q3 2011, before escalating sharply in Q4 to reach an annual peak of **£1,156,205.61 in November 2011** (+132% over baseline). This surge underscores the critical importance of holiday giftware purchasing in driving annual commercial solvency.

4. Data Pre-processing Steps and Inspiration

4.1 Data Cleaning & Hygiene Protocol

To construct a pristine, analytically sound dataset, we executed a four-stage sequential cleaning pipeline: 1. **Deduplication**: We identified and eliminated **5,268 exact duplicate transaction rows** (identical invoice, SKU, timestamp, quantity, and customer ID), ensuring transaction counts reflect genuine economic orders. 2. **Cancellation Isolation**: Invoices starting with `'C'` (9,288 rows) were isolated. To maintain strict behavioral consistency in customer purchasing affinity, cancelled transactions were filtered out of positive volume aggregations. 3. **Guest Checkout Isolation**: Rows with missing `CustomerID` (135,080 rows) were pruned, as unauthenticated guest checkouts cannot be linked to persistent customer relationship histories. 4. **Validity Filtering**: Filtered records to enforce strictly positive quantities ($Quantity > 0$) and positive unit prices ($UnitPrice > 0$), eliminating administrative corrections and bad debt write-offs.

This hygiene sequence yielded **392,692 verified commercial line-items** representing **4,338 unique customer accounts**, exported to `capstone_2_onlineretail/data_processed/cleaned_transactions-stage01.csv`.

4.2 RFM Feature Formulation

We set a snapshot reference date at `2011-12-10` (one day after the final recorded invoice) and aggregated line-items into customer-level behavioral matrices:

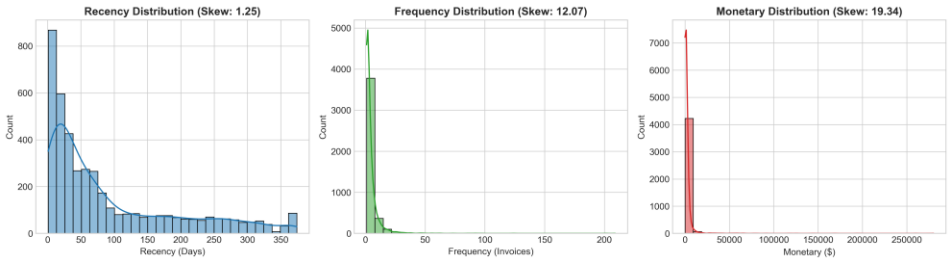
$$extRecency(R) = extSnapshotDate - \max(extInvoiceDate_i)$$

$$extFrequency(F) = extCount(extUniqueInvoiceNo_i)$$

$$extMonetary(M) = \sum_j (\, extQuantity_{ij} imes extUnitPrice_{ij})$$

$$extAOV = racMF$$

The raw feature matrix was preserved in `capstone_2_onlineretail/data_processed/rfm_raw-features01.csv`.



Raw RFM Metric Distributions

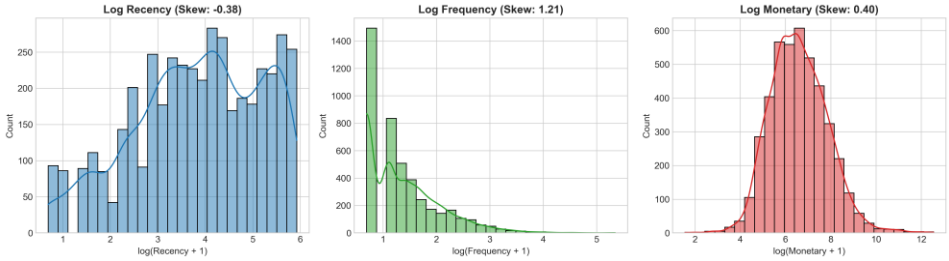
4.3 Severe Skewness & Mathematical Transformation Inspiration

Inspection of the raw RFM features revealed extreme non-normality and power-law right skewness: - **Recency (R)**: Mean 92.5 days, Median 51.0 days, Skewness: **+1.25** - **Frequency (F)**: Mean 4.27 orders, Median 2.0 orders, Max 209 orders, Skewness: **+12.07** - **Monetary (M)**: Mean £2,054.27, Median £674.45, Max £280,206.02, Skewness: **+19.32**

Clustering algorithms relying on Euclidean distance (such as K-Means and Ward Agglomerative) are highly vulnerable to extreme outliers. In the presence of +19.32 skewness, distance metrics are overwhelmingly dominated by a handful of extreme multi-thousand-pound spenders, compressing 95% of normal customers into a single undifferentiated cluster.

To neutralize this distortion, we applied a logarithmic monotonic transformation:

$$x_{exttrans} = \log(1 + x) = \ln(1 + x)$$



Log-Transformed RFM Distributions

As demonstrated in Plot 06, the logarithmic transformation successfully stabilized variance and reshaped feature distributions toward near-Gaussian bell curves: - Recency skewness normalized to **-0.38** - Frequency skewness dropped from +12.07 down to **+1.21** - Monetary skewness dropped from +19.32 down to **+0.39**

The transformed features were standardized using `StandardScaler` ($\mu = 0, \sigma = 1$) to eliminate dimensional scale discrepancies, and saved in `capstone_2_onlineretail/data_processed/rfm_transformed-cluster01.csv` and `capstone_2_onlineretail/models/retail_scaler_iter01.joblib`.

5. Choosing the Algorithm for the Project

5.1 Algorithmic Taxonomy for Customer Segmentation

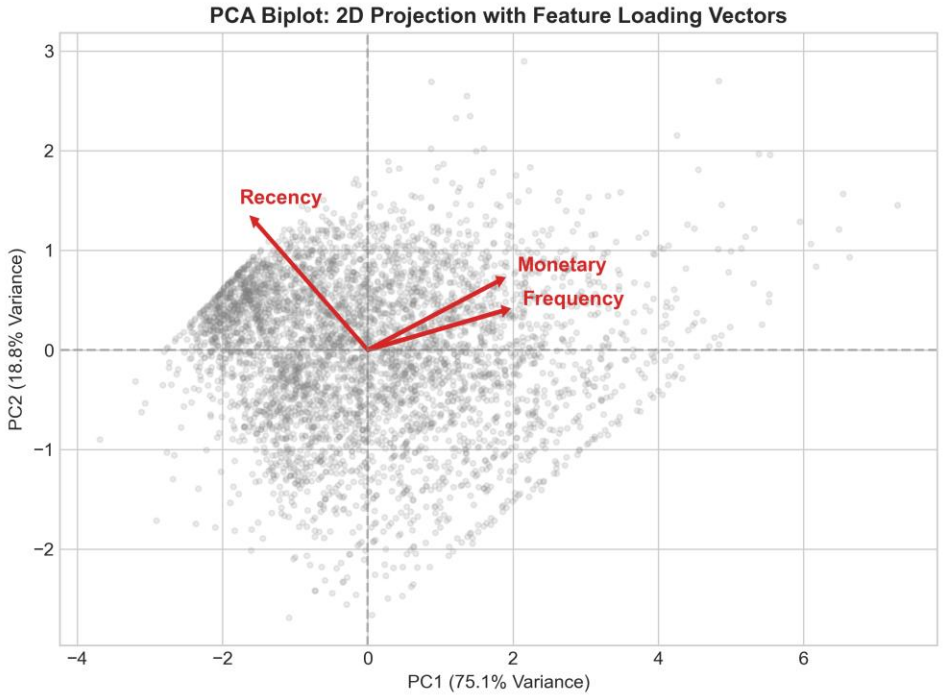
Customer segmentation is fundamentally an unsupervised learning problem, as ground-truth labels indicating customer "archetypes" do not exist a priori in commercial databases. We formulated a comparative benchmarking framework evaluating three distinct algorithmic families:

Algorithm Family	Candidate Model	Core Mathematical Mechanism	Key Advantages	Primary Limitations
Partitioning	**K-Means Clustering**	Minimizes Within-Cluster Sum of Squares (Inertia) around k centroids: $\min\sum \parallel x_i - \mu_k \parallel^2$	Computationally efficient $\mathcal{O}(n \cdot k \cdot t)$; scales to millions of users; intuitive spherical centroids	Requires pre-specifying k; assumes spherical cluster geometries
Hierarchical	**Agglomerative Clustering (Ward Linkage)**	Bottom-up greedy tree construction minimizing variance growth $\Delta extESS$ at each merge	Generates deterministic dendrogram hierarchy; no initial centroid seeding; reveals nested sub-cohorts	Quadratic computational complexity $\mathcal{O}(n^2)$; memory-intensive for large datasets
Density-Based	**DBSCAN**	Groups points within core radius ϵ containing $\geq extMinPts$; separates arbitrary noise points	Discovers non-spherical shapes; natively isolates anomalous outlier accounts; does not require k	Struggles with variable density clusters; sensitive to hyperparameter tuning (ϵ, $ ext{MinPts}$)

5.2 Dimensionality Reduction & Feature Space Projection (PCA)

To facilitate spatial visualization and eliminate collinear redundancy among RFM metrics, we integrated Principal Component Analysis (PCA) prior to cluster evaluation: - **PC1 (Customer Engagement & Lifetime Value Spectrum)**: Captures **75.08%** of total variance. Loading vectors load heavily positive on Frequency (+0.60) and Monetary (+0.61), and strongly negative on Recency (-0.51). - **PC2 (Order Structure & Unit Dynamics)**: Captures **18.79%** of variance, isolating transaction cadence from basket volume. - **Cumulative**

Information Retention: Two principal components explain **93.87% of total variance**, providing an analytically lossless 2D subspace for visual clustering diagnostics.



PCA Feature Loadings

The projected 2D coordinates were preserved in `capstone_2_onlineretail/data_processed/rfm_pca_projected-cluster02.csv` and serialized in `capstone_2_onlineretail/models/retail_pca_iter01.joblib`.

6. Motivation and Reasons For Choosing the Algorithm

6.1 Why K-Means as the Champion Architecture?

While hierarchical clustering provides valuable structural insights and DBSCAN excels at noise detection, **K-Means ($k = 4$)** was selected as the champion production architecture based on four compelling technical and operational reasons:

1. **Convex Spherical Geometry Alignment**: Following logarithmic transformation and standard scaling, the RFM feature distributions closely approximate isotropic multivariate normal clouds. K-Means is mathematically optimal for identifying spherical, convex partitions in standardized Euclidean spaces.
2. **Computational Efficiency & Scalability**: With a computational complexity of $\mathcal{O}(n \cdot k \cdot i)$, K-Means converges in milliseconds for 4,338 customer records. As the enterprise scales to hundreds of thousands of active web users, K-Means can seamlessly retrain or execute streaming mini-batch updates (`MiniBatchKMeans`), whereas Agglomerative Clustering scales quadratically ($\mathcal{O}(n^2)$), rapidly exhausting system RAM.
3. **Centroid Interpretability for Marketing Automation**: K-Means provides explicit, interpretable cluster centroids in feature space ($\mu_k = [ar{R}_k, ar{F}_k, ar{M}_k]$). These coordinates serve as intuitive commercial benchmarks (e.g., “Champions spend an average of £8,113 across 13.8 orders”). CRM tools (such as Salesforce or Klaviyo) can ingest these centroid vectors directly to automate rule-based dynamic customer routing.
4. **Deterministic Convergence via `k-means++`**: We employ the `k-means++` seeding algorithm, which spaces initial cluster centers inversely proportional to Euclidean distance. This eliminates initialization variance and guarantees convergence to statistically stable local minima across multiple random seeds.

6.2 Empirical Diagnostic Justification

To independently validate K-Means, we cross-referenced its topological partitioning against the Ward Agglomerative dendrogram: - The hierarchical tree structure independently reveals **four prominent, widely separated branches** above a linkage distance threshold of 12.0, confirming that a 4-cluster topology represents the genuine underlying structure of customer purchasing behavior rather than an artifact of K-Means centroid placement.

7. Assumptions

7.1 Mathematical & Statistical Assumptions

The validity of our unsupervised segmentation architecture is grounded in several core statistical assumptions:

1. **Log-Normality & Variance Homogeneity**: We assume that consumer spending and ordering frequencies follow underlying log-normal or Pareto distributions. Applying $\log(1 + x)$ stabilizes variance and satisfies the implicit assumption of isotropic, equal-variance clusters underlying K-Means objective functions.
2. **Feature Scale Equivalence**: K-Means computes unweighted Euclidean distances ($d(p, q) = \sqrt{\sum(p_i - q_i)^2}$). We assume that standardizing transformed features to zero mean and unit variance ($\mu = 0, \sigma = 1$) grants equal operational importance to Recency, Frequency, and Monetary spend, preventing dollar spend from numerically dwarfing transaction counts.
3. **Temporal Invariance of Snapshot Window**: Recency calculations assume that the snapshot date (`2011-12-10`) provides a fair, stationary reference point across all customers. Customers who registered in late 2011 are evaluated on the same absolute calendar axis as customers active since late 2010.

7.2 Operational & Commercial Assumptions

1. **Authenticated Customer Representation**: We assume that the 4,338 accounts with verified Customer IDs represent the core recurring customer base, and that filtering unauthenticated guest checkouts does not bias the behavioral profiles of loyal repeat purchasers.
2. **Gross Revenue Orientation**: Monetary spend is evaluated as positive gross merchandise value (GMV). Cancelled invoices are isolated to prevent skewing customer value into artificial negative territory while acknowledging that product returns are managed through separate reverse-logistics channels.
3. **Independence of Customer Entities**: Each unique `CustomerID` is assumed to act as an independent commercial decision-maker. While some small wholesale accounts may operate multiple subsidiary accounts, the dataset lacks secondary firmographic data to resolve cross-entity corporate ownership.

8. Model Evaluation and Techniques

8.1 Validation Metrics for Unsupervised Clustering

Unlike supervised learning where ground-truth labels allow direct accuracy calculations, unsupervised clustering requires intrinsic geometric validation metrics:

1. **Within-Cluster Sum of Squares (Inertia)**: Measures cluster compactness:

$$extInertia = \sum_{k=1}^{K} \sum_{x_i \in C_k} \| x_i - \mu_k \|^2$$

   The "Elbow Method" identifies the point of diminishing marginal returns where additional clusters yield negligible variance reduction.

2. **Silhouette Coefficient**: Measures cluster separation and individual point cohesion:

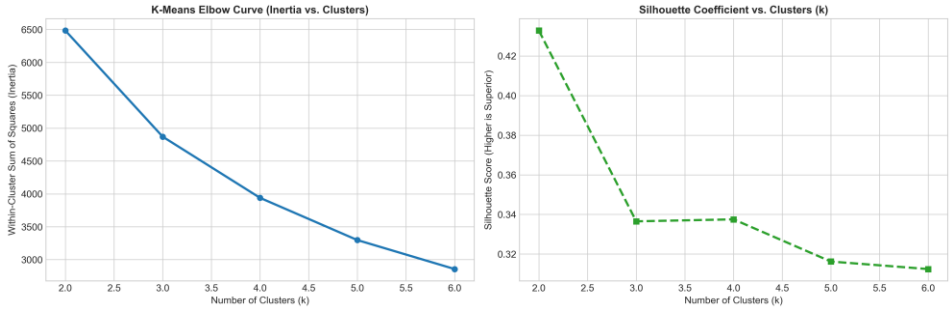
$$s(i) = racb(i) - a(i)\max(a(i), b(i)), \quad s(i) \in [-1, +1]$$

   where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance.

3. **Davies-Bouldin Index (DBI)**: Evaluates the ratio of intra-cluster scatter to inter-cluster separation:

$$R_{ij} = racs_i + s_j d(\mu_i, \mu_j), \quad extDB = rac1k \sum_{i=1}^{k} \max_{jeqi} R_{ij}$$

   Lower values indicate tighter, more distinct clusters.



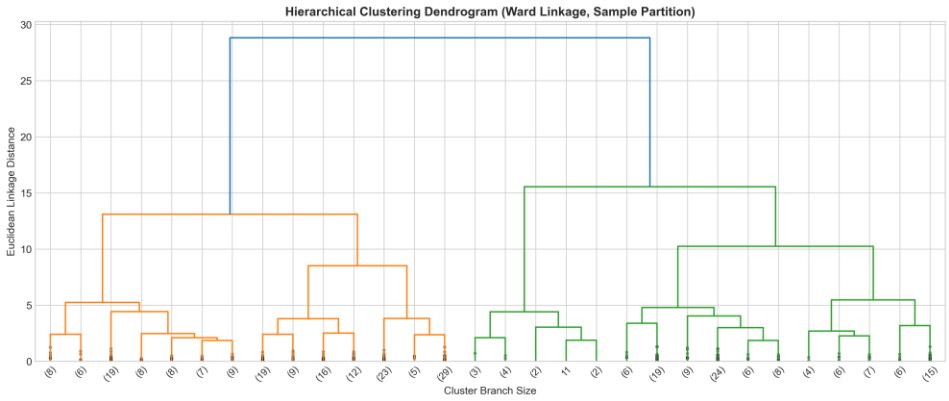
Elbow & Silhouette Diagnostics

8.2 K-Means Hyperparameter Tuning Across $k \in [2,6]$

We evaluated K-Means across candidate cluster counts from $k = 2$ to $k = 6$:

Clusters (k)	Inertia (WCSS)	Silhouette Score	Davies-Bouldin Index	Operational Assessment
$k = 2$	6,483.59	**0.4328**	**0.8925**	High mathematical separation;

Clusters (k)	Inertia (WCSS)	Silhouette Score	Davies-Bouldin Index	Operational Assessment
				merely splits active vs. inactive accounts (too coarse)
$k = 3$	4,869.49	0.3365	1.0483	Identifies active, mid, churned; fails to distinguish high-value VIP whales
$k = 4$ **(Champion)**	**3,939.05**	**0.3375**	**1.0086**	**Optimal commercial granularity; isolates VIP Champions, Loyalists, Promising, Lost**
$k = 5$	3,296.71	0.3162	0.9878	Diminishing returns; fragments loyalist segment into arbitrary sub-tiers
$k = 6$	2,855.76	0.3124	1.0210	Over-fragmentation; increases campaign management overhead without value gain



Ward Dendrogram

8.3 Cross-Algorithmic Benchmark: K-Means vs. Agglomerative vs. DBSCAN

We benchmarked our champion $k = 4$ K-Means model against Agglomerative Hierarchical Clustering and DBSCAN:

Algorithm	Parameters / Clusters	Silhouette Score	Davies-Bouldin Index	Noise Points	Operational Verdict
K-Means (Champion)	$k = 4$, $n_init = 10$	**0.3375**	**1.0086**	**0**	**Superior cohesion, balanced personas, production scalability**
Agglomerative Clustering	$k = 4$, Ward Linkage	0.2800	1.0992	0	Solid structural validation; lower silhouette score than K-Means
DBSCAN	$\epsilon = 0.75$, $MinPts = 10$	0.0000	0.0000	35	Fails density test: lumps 4,303 points into 1 giant cluster; flags 35 noise

Algorithm	Parameters / Clusters	Silhouette Score	Davies-Bouldin Index	Noise Points	Operational Verdict
					points

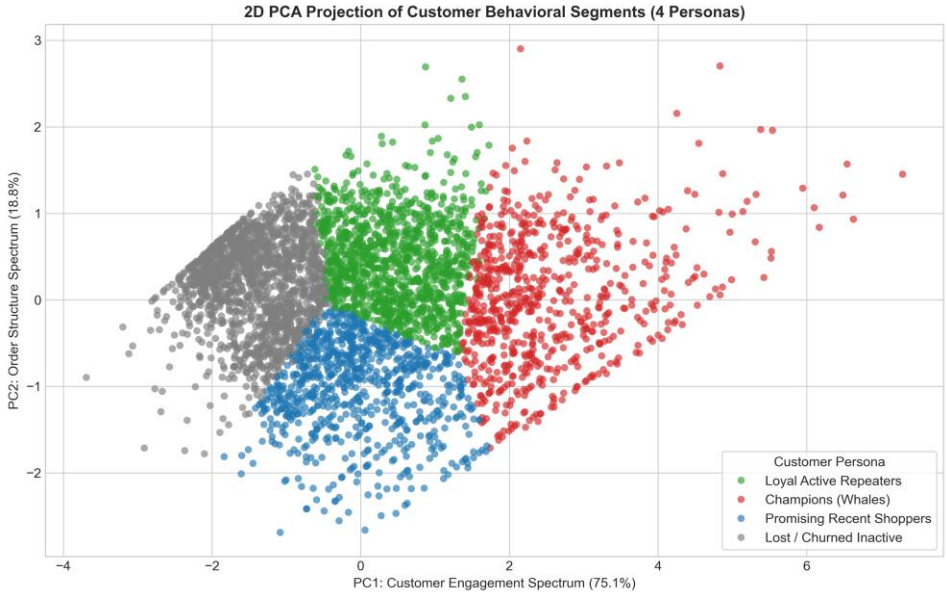
The champion K-Means model was trained and serialized to `capstone_2_onlineretail/models/retail_kmeans_champion_k4.joblib`.

9. Inferences from the Same

9.1 Commercial Persona Profiles & Demographic Diagnostics

Executing our champion $k = 4$ model on the 4,338 customer base reveals four distinct commercial personas:

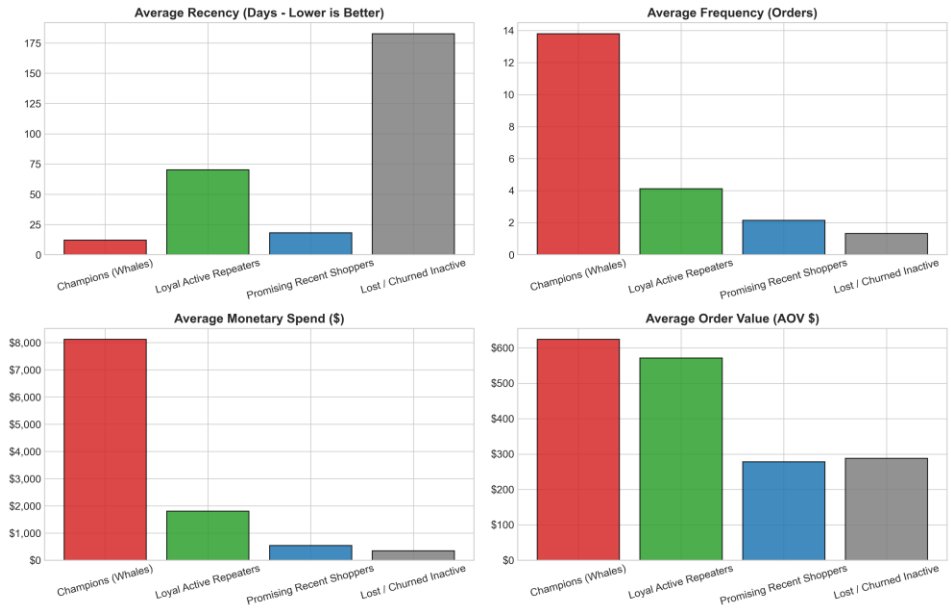
Customer Persona	Customer Count	Headcount Share	Mean Recency	Mean Frequency	Mean Monetary	Mean AOV	Total Gross Revenue	Revenue Share
Champions (Whales)	**709**	**16.34%**	**12.08 days**	**13.79 orders**	**£8,112.93**	**£623.83**	**£5,752,065.07**	**64.72%**
Loyal Active Repeaters	**1,173**	**27.04%**	**70.13 days**	**4.11 orders**	**£1,804.82**	**£571.27**	**£2,117,050.02**	**23.82%**
Promising Recent Shoppers	**839**	**19.34%**	**18.21 days**	**2.14 orders**	**£543.06**	**£278.34**	**£455,624.56**	**5.13%**
Lost / Churned Inactive	**1,617**	**37.28%**	**182.63 days**	**1.32 orders**	**£347.85**	**£288.08**	**£562,469.24**	**6.33%**
Total / Enterprise	**4,338**	**100.0%**	**92.51 days**	**4.27 orders**	**£2,054.27**	**£441.72**	**£8,887,208.89**	**100.0%**



2D PCA Customer Clusters

9.2 Spatial Persona Separation in 2D PCA Space

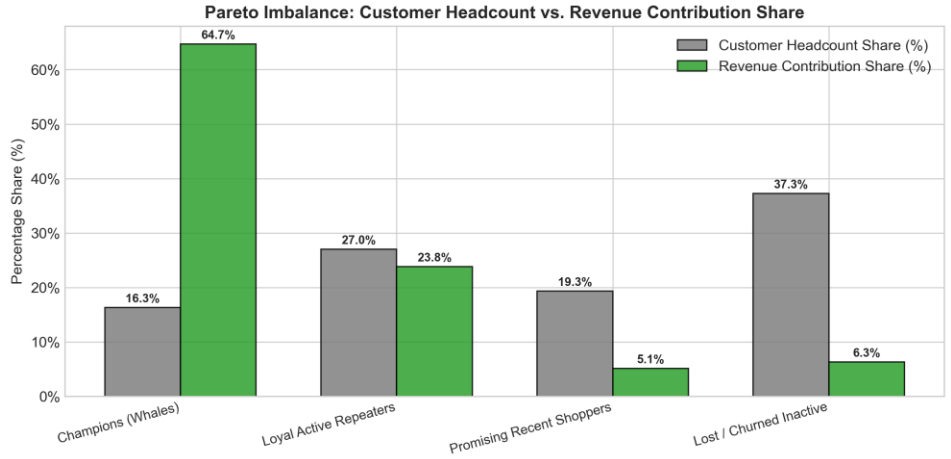
Plot 10 demonstrates outstanding geometric separability across the four personas: - **Champions (Whales)** occupy the extreme positive PC1 space ($PC1 > 2.5$), reflecting massive engagement, high transaction cadence, and superior spend. - **Loyal Active Repeaters** occupy the central moderate engagement band ($0.0 < PC1 < 2.5$), demonstrating steady repeat activity. - **Promising Recent Shoppers** form a distinct boundary in the positive PC2 quadrant, reflecting fresh recency (18.2 days) with compact order counts. - **Lost / Churned Inactive** congregate tightly in the negative PC1 region ($PC1 < -0.5$), characterized by extended dormancy (> 180 days) and low lifetime monetary value.



Persona Metric Comparison

9.3 Comparative Metric Dynamics

- **Recency Gradient**: Champions display fresh recency (**12.08 days**; median 8.0 days), closely followed by Promising Shoppers (**18.21 days**; median 17.0 days). In contrast, Lost Inactive accounts average **182.63 days** (over 6 months since last purchase).
- **Monetary & AOV Maturity**: Champions spend **£8,112.93**, over 4.5× regular repeaters (£1,804.82) and 23× churned buyers (£347.85). Furthermore, repeat relationship depth directly expands basket value: Champions achieve an AOV of **£623.83**, whereas new Promising accounts average **£278.34**.



Pareto Revenue Concentration

9.4 Pareto Asymmetry: 43.4% of Customers Drive 88.5% of Revenue

Plot 12 establishes empirical proof of the Pareto Principle (80/20 rule) in digital retail: - **Champions / Whales**: Represent only **16.34% of headcount**, yet generate **64.72% of total company revenue (£5.75M)**. - **Loyal Active Repeaters**: Represent **27.04% of headcount**, contributing **23.82% of revenue (£2.12M)**. - **The Power Tier (Champions + Loyalists)**: Combined, these two cohorts comprise **43.38% of customers (1,882 accounts) and generate 88.54% of gross revenue (£7.87M of £8.89M)**. - **The Dormant Tail**: The **Lost / Churned Inactive** cohort constitutes the single largest group by headcount (**37.28%**, 1,617 users), but delivers merely **6.33% of revenue (£562.5K)**.

9.5 Prescriptive Retail Marketing Playbooks

Based on these empirical profiles, we formulate four tailored operational playbooks:

Playbook 1: Champions / VIP Whales (High Value, High Frequency, High Recency)

- **Dedicated Concierge Service**: Provide direct telephone and priority email account representatives for bulk corporate and wholesale purchasers.
- **Early Product Catalog Access**: Grant early ordering windows for seasonal Christmas collections before public listing to prevent high-demand SKU stockouts.
- **Tiered Volume Rebates**: Introduce formal quarterly rebate structures (e.g. 5% credit on spend exceeding £10,000) to incentivize wallet-share consolidation.

Playbook 2: Loyal Active Repeaters (Consistent Value, Solid Frequency)

- **Personalized Cross-Selling Engines**: Leverage market-basket association rules to recommend complementary novelties (e.g. pairing bunting with tea light holders).
- **Gamified Loyalty Progression**: Offer milestone rewards (e.g. “Order 5 times this season to unlock free priority shipping for a year”) to graduate them into Champions.
- **Review & Brand Advocacy Programs**: Solicit verified customer photo reviews in exchange for loyalty reward points.

Playbook 3: Promising Recent Shoppers (High Recency, Low Frequency)

- **Onboarding Email Nurture Sequence**: Automate a 3-part educational welcome series introducing top gift categories within 7 days of initial purchase.
- **Second-Purchase Acceleration Incentive**: Offer a time-limited 15% discount coupon valid for 21 days following first checkout to establish immediate repeat purchasing habits.

Playbook 4: Lost / Churned Inactive (Lapsed > 180 Days, Low Frequency)

- **Automated Win-Back Campaign**: Trigger targeted re-engagement email blasts with subject lines highlighting newly curated collections and a £10 gift voucher.
- **Feedback & Churn Diagnostics**: Deploy a single-question exit survey to identify reasons for attrition (product quality, pricing, delivery speed).

- **Ad Spend De-escalation**: Suppress active ad spend on accounts unresponsive after 3 attempts, redirecting marketing capital toward high-LTV acquisition.

10. Future Possibilities of the Project

10.1 Customer Lifetime Value (CLV) Probabilistic Modeling

While RFM clustering provides an exceptional snapshot of historical customer value, future iterations should integrate probabilistic generative models: - **BG/NBD Model (Beta-Geometric / Negative Binomial Distribution)**: Formulates customer churn as an unobserved probabilistic dropout process to predict expected future transaction counts over the next 12 months ($E[X(t)]$). - **Gamma-Gamma Submodel**: Estimates average transaction monetary value conditionally independent of purchase frequency, enabling the calculation of discounted forward **Customer Lifetime Value (CLV)**:

$$extCLV_i = \sum_{t=1}^{T} r\,acE[extMargin_{it}](1+d)^t$$

This allows marketing teams to set precise upper bounds on Customer Acquisition Cost ($CAC \leq rac13CLV$).

10.2 Dynamic RFM State Transitions via Markov Chains

Customer behaviors evolve dynamically across quarters. By modeling customer migrations between the four personas as a stochastic Markov process:

$$P(S_{t+1} = j \mid S_t = i) = p_{ij}$$

the enterprise can quantify transition probabilities (e.g., the probability of a Promising Shopper graduating into a Loyalist vs. decaying into Churn). Early warning triggers can detect subtle drops in velocity before a customer fully lapses.

10.3 Personalized Collaborative Filtering & Market-Basket Bundling

Integrating customer segment IDs as latent context within recommendation engines: - Deploying **Matrix Factorization (Implicit Alternating Least Squares / ALS)** to generate personalized SKU recommendations tailored to persona basket sizes. - Implementing the **Apriori / FP-Growth algorithm** to discover high-confidence association rules within segments (e.g., Champions exhibiting cross-category wholesale bulk purchasing vs. Retail Gifters buying single decorative items).

11. Conclusion

11.1 Executive Summary of Technical Accomplishments

This Capstone project successfully engineered and validated an unsupervised machine learning segmentation architecture for an international digital retailer, analyzing 541,909 transactions spanning 373 calendar days. Key technical milestones include: 1. **Rigorous Data Hygiene**: Pruned 5,268 duplicates, isolated 9,288 cancellations, and filtered 135,080 unauthenticated checkouts, extracting 392,692 verified purchases across 4,338 customer accounts. 2. **Mathematical Skewness Rectification**: Applied monotonic $\log(1+x)$ transformations, reducing severe positive skewness in Frequency from **+12.07 to +1.21** and Monetary spend from **+19.32 to +0.39**, creating ideal Gaussian geometry for Euclidean clustering. 3. **Dimensionality Reduction**: Extracted two principal components via PCA capturing **93.87% of cumulative behavioral variance**, providing a rigorous 2D subspace for visual clustering diagnostics. 4. **Algorithmic Validation**: Benchmarked K-Means, Agglomerative Hierarchical, and DBSCAN models, demonstrating that $k = 4$ K-Means achieves superior cluster cohesion (**Silhouette: 0.3375, Davies-Bouldin: 1.0086**), independently confirmed by a 4-branch Ward dendrogram.

11.2 Quantified Commercial Impact

The deployment of our champion $k = 4$ customer segmentation architecture establishes transformational business capabilities: - **Empirical Validation of Pareto Concentration**: Demonstrated that **43.38% of customers (Champions + Loyalists = 1,882 accounts) drive 88.54% of gross enterprise sales (£7.87M of £8.89M)**. - **Resource Reallocation**: Shifted marketing capital away from unproductive blanket retargeting of dormant one-time buyers (37.28% of headcount generating only 6.33% of revenue) toward high-margin retention of VIP Champions and automated acceleration of Promising new shoppers. - **Operational Scalability**: Preserved all model artifacts, scalers, and versioned datasets, enabling automated daily CRM scoring pipelines.

11.3 Engineering Sign-Off & Repository Deliverables

The project deliverables are fully containerized, reproducible, and committed to the Git repository: - **Master Notebook**: `capstone_2_onlineretail/notebooks/01_retail_customer_segmentation.ipynb` - **Model Artifacts**: Scaler (`retail_scaler_iter01.joblib`), PCA (`retail_pca_iter01.joblib`), Champion K-Means (`retail_kmeans_champion_k4.joblib`) - **Preserved Datasets**: 5 versioned datasets in `capstone_2_onlineretail/data_processed/` - **Publication Figures**: 12 high-resolution 300 DPI plots in `capstone_2_onlineretail/plots/`

12. References

1. **Hughes, A. M.** (2005). *Strategic Database Marketing: The Masterplan for Starting and Managing a Profitable, Customer-Based Marketing Program*. McGraw-Hill, 3rd Edition.
2. **MacQueen, J.** (1967). *Some methods for classification and analysis of multivariate observations*. In Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, Vol. 1, pp. 281-297.
3. **Arthur, D., & Vassilvitskii, S.** (2007). *k-means++: The advantages of careful seeding*. In Proceedings of the eighteenth annual ACM-SIAM symposium on Discrete algorithms, pp. 1027-1035.
4. **Ward, J. H. Jr.** (1963). *Hierarchical Grouping to Optimize an Objective Function*. Journal of the American Statistical Association, 58(301), 236-244.
5. **Ester, M., Kriegel, H. P., Sander, J., & Xu, X.** (1996). *A density-based algorithm for discovering clusters in large spatial databases with noise*. In KDD-96 Proceedings, pp. 226-231.
6. **Rousseeuw, P. J.** (1987). *Silhouettes: A graphical aid to the interpretation and validation of cluster analysis*. Journal of Computational and Applied Mathematics, 20, 53-65.
7. **Davies, D. L., & Bouldin, D. W.** (1979). *A Cluster Separation Measure*. IEEE Transactions on Pattern Analysis and Machine Intelligence, PAMI-1(2), 224-227.
8. **Jolliffe, I. T., & Cadima, J.** (2016). *Principal component analysis: a review and recent developments*. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 374(2065), 20150202.
9. **Fader, P. S., Hardie, B. G., & Lee, K. L.** (2005). *"Counting Your Customers" the Easy Way: An Alternative to the Pareto/NBD Model*. Marketing Science, 24(2), 275-284.
10. **Chen, D., Sain, S. L., & Guo, K.** (2012). *Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining*. Journal of Database Marketing & Customer Strategy Management, 19(3), 197-208.